

Virus as Quasi-Living Bio-Material for Interaction Design: Practical, Ethical, and Philosophical Implications

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ABSTRACT

The interaction design research community continues to benefit from material-focused approaches, and from the diversity of materials under investigation. One category of such material is bio-materials of microbial origin, such as bacteria, mycelium, moulds, and *Euglena*. However, despite the increasing momentum towards bio-material based research, one type that is yet to be investigated in HCI, is viruses; an infectious, sub-microscopic, quasi-living, computational bio-agent. This paper initiates exploration of Human-Virus Interaction (HVI), through a material lens. This was achieved first by generating a literature-based material profile sketch of viruses, highlighting some of their distinct and/or unique material properties, characteristics, composition, and meaning. The components of the profile were then used as anchor points, to unpack the practical, ethical, and philosophical implications that are associated with viruses, and those that could be considered by researchers to help in their preparation of working with viruses in interaction design.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; *HCI theory, concepts and models*.

KEYWORDS

Virus, Bio-materials, Materiality, Human-Virus Interaction, Bio-Digital, Biological-HCI, Pathogens, Virology

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1 INTRODUCTION

Over the last two decades, recognizing the importance of a material's properties and qualities in shaping the design of interactive artefacts, an increasing proportion of HCI researchers have been conducting interaction design through a material lens.

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Such works have so far focused on a variety of material types: From traditional to modern (e.g., [53, 63, 69, 79]), analogue to digital (e.g., [10, 17, 36, 56]), artificial to organic (e.g., [4, 72, 77, 78]), and embodied to in-bodied (e.g., [47, 61, 65, 67]); an eclectic range and composites of materials have been studied and discussed.

Part of the success of this diversification has been due to the willingness to change *status quo* of interaction design. Especially those that push the boundaries on what design materials are, and how they are supposed to be perceived – concepts such as tangible bits [28], radical atoms [27], computational composites [70], and ephemeral user interfaces [19]) to name just a few – have paved fresh pathways for research.

Likewise, approaches such as inspirational bits [66], have shifted expectations on which materials could and could not be explored, by empowering designers to confront the unconventional and seemingly incomprehensible, “black-boxed” [66] (p.1561) and “immaterial” [7] (p.205) materials, in a meaningful and productive manner.

It is this spirit of additive reform, alongside an open mind, that are the basic premises underlying the thinking processes in this paper. And it invites you to do the same – temporarily at least – as it entertains “matters” that are not usually considered in design research: Viruses.

A Latin word for *poison*, virus is an infectious, sub-microscopic, computational bio-material. Comprising simply of a piece of genetic material (i.e., DNA or RNA) packaged by a layer of protein coating, its physicality is deceptively inconspicuous, as viruses continue to profoundly shape how we experience and interact with the world. Despite their significance and abundance, viruses are yet to be considered nor analysed as a design material. Using this disparity as a starting point, the paper will attempt to provide a sufficiently provocative discussion around material and materiality of viruses; narrated in the context of, and with the intention of enriching possibilities for, computational bio-material exploration within interaction design.

2 RESEARCH OPPORTUNITIES

2.1 New Bio-Material for HCI

In terms of bio-material explorations, the HCI community has seen a recent momentum in this area over the last few years.

Mostly consisting of the microbial variety, these so-called “living materials” have been presented and discussed in interaction design research through various domains. These include interactive public installations (e.g., [3, 44, 45]), hybrid computer games (e.g., [32, 34, 41]), bio-fabrications (e.g., [71, 78]), sustainable e-wearables (e.g., [55, 72]), and growable robotics (e.g., [58]), to name just some of the examples from the continually expanding portfolio.

Interestingly, most of these materials under explorations within the realms of the microbial kingdom, have primarily come from bacteria, yeasts, fungi (mycelium and moulds), and *Euglena*, the commonly recognized and familiar forms that have been domesticated, publicized, and studied in schools.

Yet as for viruses, despite their abundance and their impact on how we live, interact, and experience the world, are currently sidelined from discourse as a serious design material. It is not hard to assume the possible reasons why (e.g., inaccessibility, pathogenicity, ignorance, lack of design heuristics, and stigma associated with viruses, etc). However, this paper aims to oversee these potential barriers, and offers an initial, constructive analysis of viruses, that could lead to productive and meaningful discussions on the feasibility and implications of the agents as a new bio-material for interaction design.

This paper will talk about viruses in terms of their material and materiality. It is material-focused, and not driven or restricted by specific products that may arise from it, thus leaving future applications open ended. However, below I briefly highlight some areas for implementation to provide at least some basic context.

2.2 Human-Virus Interaction

As a material for interaction design, virus forms a part of the Human Virus Interaction (HVI) framework (Figure 1). A variation of the Human Biology Interaction (HBI) [45] (p.2593), HVI is a simplified model that describes the interactive pathways that could be formed between viruses, humans, and the digital world.

Whilst highly reductive in describing the complex and entangled nature of viruses' involvement in our digitally mediated lives, HVI is nevertheless a convenient model to infer the basic types of interactive applications that could be designed around viruses. For example, the direct physical interactions that can be formed between humans and viruses (H-V, Figure 1) indicate a design space for applications that would minimise or terminate human-virus interaction (e.g., protective wearables and therapeutics).

On the other side of the framework, digital-virus interaction (D-V, Figure 1) denotes a field of digitally assisted diagnostic tools that could be designed to sense and indicate presence of virus material in human serum and/or the environment. Furthermore, the human-digital pathway (H-D, Figure 1) points towards a field of user-driven software and interface development, for appliances that deal with viruses "in the wild".

The point being, considering viruses as material for interaction design opens up a rich design and research space with a variety of stakeholders. It could lead to a diverse set of timely applications within the realms of medicine, agriculture, and possibly beyond; that are able to protect from, eradicate, heal, sense, diagnose, and even facilitate our relationships with viruses.

3 OBJECTIVES

Despite the compelling outlook for considering viruses as material for interaction design, given its novelty as a research subject in HCI, there is currently no frame of reference for constructive exploration in the community. This paper aims to provide such reference, in the form of practical, ethical, and philosophical implications. Extrapolated from a preliminary assessment of the material and materiality

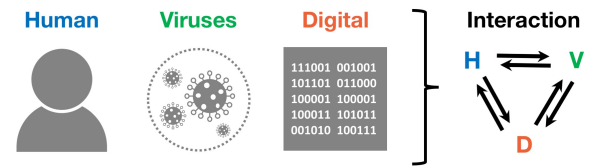


Figure 1: Human-Virus Interaction Framework

of viruses, these implications could be addressed and/or considered by prospective researchers who wish to work with viruses in the context of interaction design and HCI.

4 MATERIALS AND METHODS

4.1 Material Profile Sketch

Information on various material aspects of viruses were gathered from literature in virology, molecular biology, medicine, sociology, and anthropology. These aspects were divided into the following categories: material property, character, composition, and meaning, as defined and described in Wiberg's 2014 publication, *Methodology for Materiality: Interaction Design Research Through a Material Lens* [81]. Other aspects mentioned by Wiberg (i.e., material appearance, authenticity, aesthetics, and quality) were omitted (for the time being), as they offered too much granularity for the scope of this paper. The idea was to introduce virus as a topic for initial material discussion, which can then be used as pivots for unpacking their initial set of implications in interaction design.

Likewise, consultations had been made on other existing material profiling frameworks. However, while they continue to offer valuable grounding in interaction design, they proved to be impractical as a template, but only for the limited scope of this paper. For example, some offered too high level of abstraction unsuitable for the sketching phase of material analysis (e.g., [29]), while others focused on aspects of materials (e.g., experiential [25] and application-specific [18]) that are most relevant for researchers that have a finished interactive product in mind.

4.2 Sketch Analysis

The components of the material profile sketch were subsequently analysed and discussed, in the context of providing points for building future methodological frameworks that could be utilized for interaction designers.

5 VIRUS MATERIAL PROFILE SKETCH

5.1 Properties and Character

In terms of physicality, virus sizes typically range between 20 to 350 nanometres [26]. At the extreme ends beyond this range, are porcine circoviruses, with the smallest diameter of 17 nm [68], and *Pithovirus sibericum* with the largest observed diameter of 1,500 nm [46]. Viruses are invisible; Also odourless, tasteless, soundless, and non-tactile. The point is, that viruses are imperceptible to the unaided human sensory organs. Unless of course, should the viruses infect a living cell and start replicating themselves by hijacking the host's cellular machinery. Only the visible and physical symptoms

expressed by the host can mark the viruses' presence, enabling tangibility of the pathogens to rise to the fore.

Bio-chemically speaking, at the core of the virus reside a string of genetic material, in the form of either DNA (deoxyribonucleic acid) or RNA (ribonucleic acid), that encode vital instructions designed to ensure viruses' survival. These codes may exist as a single stranded entity or as a double stranded pair. In relation to their genetic contents, viruses are a computational material, a widely-accepted concept that has existed in life sciences for several decades [51]. Upon an infection, viruses make molecular computations using a combination of DNA, RNA, and protein molecules, to decide whether to attack and kill the host, or to remain dormant within it [51]. Moving on to examining the viruses' outer edges that encase the DNA/RNA, is a layer of molecular coating, which always is comprised of proteins, and sometimes also an extra layer of lipids (fat molecules) that are collectively known as "the envelope".

As for their biological attributes, I outline just two of their hallmarks here. First, is the pathogenic capability of viruses. The mechanisms with which they achieve pathogenesis are complex and varied, involving multiple enzymes and biochemical pathways (e.g., necrosis [30], apoptosis [2], and replication inhibition [38] of the host cells). But essentially, viruses are designed to disrupt the host's normal cellular functions, all programmed and initiated by the invaders' genetic blueprint material.

Second, viruses are not designed for independent living. That is, viruses cannot replicate on their own, but have to rely on the resources of the victims of their invasion in order to achieve it. This dependency has been the crux of most arguments in life sciences community, the arguments that reject the motion of viruses as living organisms (e.g., [9, 75]).

5.2 Composition and Meaning

Viruses are the most abundant biological entities in the biosphere [20], and were the important drivers of evolution both in microbes and humans [22, 37]. A major facet of viruses' ecology of its ancient past and the present, is a parasitic relationship between them and their host species. Such relationship is not necessarily immediately destructive either, as viruses are also able to form dormant composites with the host. This is often achieved by embedding part of the viral DNA within the host's genome [31], effectively transforming the genetic identity of the host.

In terms of its social and cultural meaning, viruses have been mostly associated with negativity. In particular, viruses have often been branded as "enemies of life" [73] (p.489), mostly in the media (e.g., [15, 49, 62]). As Villareal writes: "*our history and suffering has led us to view viruses as the hidden enemy, evil entities that simply seek to destroy life (...)*" [74] (p.14).

Similarly, as a metaphor, a computer "virus" [13] is used to describe malware that can comprise the normal performance of a computer, while the term had also been used as a derogatory quip, a political contagion [76] (p.179). An exception that do not project immediate negativity is the recent use of the term "viral" to describe a digital artefact (e.g., a social media post or an online video) that gain popularity within a short time period [80].

Furthermore, the negative connotations of viruses have a composite effect, meaning that the negativity is not isolated to the

viruses themselves, but extend beyond, to those who are infected by them. Social stigma experienced by those who suffer from viral infections have been well documented (e.g., [6, 14, 16, 64]).

6 DISCUSSION

The following outlines and discusses the implications that could be drawn from the material profile sketch of viruses. The idea is to use these as reference points for the development of a methodological framework that would facilitate the exploration of viruses, as well as similar bio-materials. For better clarity and coherence, the implications have been categorized as one of three types: practical, ethical, and philosophical. These are discussed below.

6.1 Practical Implications

Hands-off Strategy. Given the pathogenic nature of many viruses, the "hands-on" approaches of crafts that are often adopted in interaction design (e.g., [10, 23]) are not applicable. As an example, while direct skin contact (and even smell) may be necessary for quality control of certain technical processes (e.g., sanding wood, odour removal from leather, etc.), this is not possible with viruses. For professionals who work in Biosafety Level 4 (BSL-4) laboratories that facilitate the study of some of the pathogenic types (e.g., Ebola), spatio-temporal distance is created between the investigator and the material (figure 2). These are achieved by means of 1) special "HAZMAT" suits (with its own air supply system), 2) airlocks, and 3) rigorous equipment safety and decontamination protocols pre and post experimentation.



Figure 2: Dr. Judith Olejnik performing BSL-4 work at Boston University's National Emerging Infectious Diseases Laboratories. (Image credit: Dr. Elke Mühlberger).

Consequently, designers may need to consider using additional technologies, such as remotely controlled robots, for hands-off material handling. Further still, iterative design and material modifications that form integral parts of established research methods (e.g., research through design and design thinking), may not be practical, forcing the designer to find an alternative and/or a new research strategy.

New Collaborators. Working with viruses will require collaborators from at least two new disciplines that currently remain largely unfamiliar to interaction design. First is virology. While the HCI community have so far witnessed increasing collaborative research between interaction designers and those from various branches of microbiology – namely bacteriology (e.g., [3, 39, 40]) and mycology (e.g., [42, 48, 78]) – projects that cross-pollinate with virology have not yet been materialized. The pheno and genotypic idiosyncrasies that distinguish viruses from other microbial lifeforms inevitably change the protocols involved for study, as well as research agenda of those that work with them. And in turn, the resulting social dynamics that are formed between designers and virologists, so far remain unknown.

The second new potential discipline for collaboration is synthetic biology. The DNA-driven, computational nature of viruses have naturally historically invited those who design, engineer, and standardize genetic circuits, i.e., synthetic biologists, who pursue the development of genetically engineered antiviral solutions (e.g., [1, 5]). For interaction designers to understand and explore viruses on a purely bio-computational level, understanding the language of molecular biology would be appropriate. Treating viruses as computers that require biological programming for its modification and/or for the development of other computers to interact with them, would warrant such insight. However, the gap that exist in HCI, in terms of accepting and meaningfully integrating molecular biology as transferable methodology for interaction design remain unabridged, and would require further interdisciplinary dialogues.

Longer Timeframe. The biological nature of viruses as a design material means that their temporal characteristics in response to external stimuli (i.e., an input) is significantly different to those from non-biological materials. The biological response will appear slower than digital responses, as biochemical reactions necessary to produce visible outcomes (e.g., protein aggregates) occur on a molecular level. In HCI, this phenomenon is widely accepted and expected, with many recognizing the potential of slowness as a useful design tool that could be harnessed to improve user/player experiences (e.g., [33, 35, 40, 57]).

In addition, from a bureaucratic standpoint, working with viruses will require relatively longer preparation time in contrast with those involved with non-dangerous counterparts. Safety training, safety test, and paperwork necessary to grant designers the permit to work alongside with virologists in a controlled environment will need to be factored into their project scheduling.

6.2 Ethical Implications

Therapeutic Approach. Studying viruses is an inexorably destructive process. As viruses naturally display their material characteristics through the host, studying such characteristics will inevitably lead to harm and/or loss of life for the host cell. As such, material studies involving viruses is a morally contestable practice, at least when one is placed under the premise of striving towards respect and preservation of living forms. This ethical dimension is absent in methods of (non-living) material study that are currently being implemented in interaction design. In crafts, electronics, and programming, the intensities of moral consequences that result from

destruction of a material, is arguably relatively low. In fact, in some cases, destruction of material is encouraged (e.g., [21, 50]).

With this in mind, I propose the concept of therapy as an alternative *modus operandi* for material study. This could be applied not only for viruses, but other parasitic pathogens that achieve tangibility at the expense of another living entity. At minimum, it is a non-destructive material study approach, and at best it can become a preservative or even an additive form. The idea is to orientate the material exploration with a broader aim to heal the material under the study. This may be achieved by collaborating with experts whose research agenda is in close alignment with the therapeutic approach proposed here (e.g., those related to vaccine, diagnostic, and therapeutics development groups).

Material Neutralization. One of the strategies currently used by scientists to study viruses is the creation of “pseudotype viruses” (e.g., [8]) – a sort of artificially generated viral chimera, which consists of fused elements originating from two different strains. The result is neutralization from pathogenicity, enabling researchers to study the virus under less restrictive laboratory conditions. Pseudotypes are therefore a compromise, lessening the purity and essence of the material. They are also a transformational composite, altering the material identity of the original form. Whether this is a reasonable or an interesting compromise, from an interaction design point of view, is a question worth considering for future designers.

6.3 Philosophical Implications

Unbiased Approach. Given the deluge of negativity surrounding viruses as a whole, they make for an odd and risky material for research. Designers may feel ostracised or discouraged by their community for expressing their intentions of working with viruses. Currently, as there are no documented cases within HCI, of material-centred research involving extreme materials of such nature and magnitude, it is difficult to identify an existing guideline to address this potential social problem. It is also a challenge that the community should discuss through open dialogue.

Non-Humanist Imperative. One of the well-recognized global phenomena that help perpetuate the overwhelming negative perception of viruses, is what Wald calls *The Outbreak Narrative* [76]. It is a term used to describe how societies talk about, and obsess over, viral epidemics. It is comprised of three plot lines, which are: 1) the emergence of a viral infection, 2) the spread of the infection globally through various networks, and 3) the containment (and possible eradication) of the virus through epidemiologic work. And as I write this paper, these plot lines have been played, and are being played out with *Covid-19* within schools, governments, businesses, through various media outlets, and amongst family and friends.

While the narrative can serve vital humanitarian purposes – saving lives, preserving the economy, protecting vulnerable and future generations, etc – there is also a potential risk of the predominant focus over pathogenic viruses, in overshadowing or limiting discourse around viruses that are non-pathogenic and potentially useful (e.g., human virome [43] and bacteriophages [11]). With this in mind, I highlight a possibility for a post-outbreak narrative in interaction design, one that need further unpacking and fine-tuning

over time; It is a form of non-humanist, “more-than human” design approach towards exploring viruses (and indeed other pathogenic materials); seeing beyond the danger posed by the material, embracing their potential redeeming qualities, and to explore their material aspects in the purest sense possible.

Defining “Life” in HCI. As explained in the material analysis, viruses are considered quasi-living, with their status as living organism being contested in fields of life sciences. In such communities, viruses had been historically and mostly regarded as non-living. However, a publication in *Nature* in 2008 have provided the latest argument contesting this motion [60]. It presented a case of a virus strain that can become “ill”, upon attacks from virophages (viruses that attack other viruses). This ability for viruses to fall ill was then used as the basis for arguing that viruses are a living form, the rationale which has surprised many in the science field (e.g., [24]).

As Forterre [24] pointed out, the decision which labels viruses as living or non-living, is ultimately a philosophical one, with no correct criteria to achieve it. In HCI, how we define viruses in terms of its living status is unknown. And perhaps more pressingly, to my best knowledge, the community has not yet clarified on how life is described, nor what living things constitute. This may be a timely opportunity, collectively as the HCI community, to call for further clarity, at least from the perspective of deciding whether viruses should be considered a living material in the context of interaction design. This is a significant decision, as it would 1) shape ethical discourse and ethical frameworks that could be developed in the future, and 2) influence the choice of existing frameworks under which interactive systems should be evaluated. For example, HCI-specific premises such as *Empathetic Living Media* [12], *Species-Appropriate Computer Mediated Interaction* [52], *Living Bits* [59], and *Living Media Interfaces* [54], etc., can only operate meaningfully with the premise of living status of the subject material. If the decision can be a philosophical one, why not develop our own HCI philosophy for defining life?

7 LIMITATIONS AND FURTHER WORK

The material profile of viruses presented in this paper is currently divisible into two levels of analysis - micro and macro - as described by Wiberg [81]; materials (properties and character) and wholeness (material composition and meaning). While these two levels provide sufficient material for provoking initial discussions on viruses as design material, further granularity of levels of analysis would help towards a more rounded assessment of the biomaterial, in the long term. Viruses could be analysed further through analysis of their texture (i.e., appearance and authenticity) and details (i.e., aesthetics and quality).

In terms of the practical, ethical, and philosophical implications; while the ideas presented offer useful threads for discourse, it could be further improved by unpacking and developing them into workable research questions. For example, a useful exercise would be to compare existing methodologies involved in other, non-biological (but similarly potentially dangerous) material exploration. How are these material studies conducted, and what are the lessons that could be had for implementation in viruses; what type of assistive

technologies do they use, are there case studies of chimeric substitutions, and what kind of philosophical stances do designers employ as premises for the studies?

8 CONCLUSION

The interaction design field continues to benefit from material-focused research and the diversity of materials under investigation, including an array of bio-/living materials. This paper examined viruses through a material lens, and in the context of introducing viruses as a new bio-material for interaction design. Through the analysis of material profile generated, the paper outlined and discussed the potential practical, ethical, and philosophical implications that are worth considering, for those who wish to work with viruses (and other similar materials) within interaction design.

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